

24K—0559

BAZIL-UDDIN-KHAN

BSCS-1H

TASK 1).

CODE:

// 24K-0559 BAZIL UDDIN KHAN

#include <iostream>

using namespace std;

void Adddata(int * ptr, int Totalelements);

void Dma(int *& ptr, int Totalelements);

void Displayinfo(int * ptr, int Totalelements);

void Freememory(int * ptr, int Totalelements);

int main()

```
{  
    int Totalelements;  
    cout << "Enter total elements " << endl;  
    cin >> Totalelements;  
  
    int * ptr;  
  
    Dma(ptr, Totalelements);  
    Adddata(ptr,Totalelements);  
    Displayinfo(ptr,Totalelements);  
  
    Freememory(ptr,Totalelements);  
  
    return 0;  
}  
void Adddata(int * ptr, int Totalelements)  
{  
  
    for(int i =0; i < Totalelements;i++)
```

```
{  
    int num;  
    cout << " Enter number " << endl;  
    cin >> num;  
    ptr[i] = num;  
}  
}
```

void Dma(int *& ptr, int Totalelements)

```
{  
    ptr = new int[Totalelements];  
}
```

void Displayinfo(int * ptr, int Totalelements)

```
{  
    int Totalsum =0;  
    for(int i =0; i < Totalelements;i++)  
    {  
        Totalsum = Totalsum + ptr[i];  
    }
```

```

    }
    cout << "The total sum is " << Totalsum << endl;
    float average = (Totalsum/Totalelements);
    cout << "The average of array is " << average << endl;
    int Max = ptr[0];
    for(int i =0; i < Totalelements;i++)
    {
        if(ptr[i] > Max)
        {
            Max = ptr[i];
        }
    }

    cout << "Max element in array is " << Max << endl;

}

```

```

void Freememory(int * ptr, int Totalelements)
{

```

```
delete [] ptr;

}
```

OUTPUT:

main.cpp	Output
<pre>1 // 24K-0559 BAZIL UDDIN KHAN 2 #include <iostream> 3 using namespace std; 4 5 void Adddata(int * ptr, int Totalelements); 6 7 void Dma(int *& ptr, int Totalelements); 8 void Displayinfo(int * ptr, int Totalelements); 9 10 void Freememory(int * ptr, int Totalelements); 11 12 13 int main() 14 { 15 int Totalelements; 16 cout << "Enter total elements " << endl; 17 cin >> Totalelements; 18 19 int * ptr; 20 21 Dma(ptr, Totalelements); 22 Adddata(ptr, Totalelements); 23 Displayinfo(ptr, Totalelements); 24 25 Freememory(ptr, Totalelements); 26 27 return 0; 28 } 29 void Adddata(int * ptr, int Totalelements) 30 { 31 32 for(int i =0; i < Totalelements;i++) 33 { 34 int num;</pre>	<pre>Enter total elements 3 Enter number 1 Enter number 2 Enter number 3 The total sum is 6 The average of array is 2 Max element in array is 3 === Code Execution Successful ===</pre>

TASK#2).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
```

```
#include <iostream>
```

```
using namespace std;
```

```
void Adddata(int ** ptrone, int ** ptrtwo, int  
rowsone,int colsone,int rowstwo,int colstwo);
```

```
void Dma(int **& ptrone, int **& ptrtwo, int rowsone,  
int colsone,int rowstwo, int colstwo);
```

```
void Addition(int ** ptrone,int ** ptrtwo, int rowsone,  
int colsone,int rowstwo, int colstwo);
```

```
void Subtraction(int ** ptrone,int ** ptrtwo, int  
rowsone, int colsone,int rowstwo, int colstwo);
```

```
void Multiplication(int ** ptrone,int ** ptrtwo, int  
rowsone, int colsone,int rowstwo, int colstwo);
```

```
void Freememory(int ** ptrone, int ** ptrtwo,int  
rowsone, int rowstwo);
```

```
int main()
```

```
{
```

```
    int rowsone;
```

```
    cout << "Enter total rows for Array 1 " << endl;
```

```
cin >> rowsone;
```

```
int colsone;
```

```
cout << "Enter total cols for Array 1 " << endl;
```

```
cin >> colsone;
```

```
int rowstwo;
```

```
cout << "Enter total rows for Array 2" << endl;
```

```
cin >> rowstwo;
```

```
int colstwo;
```

```
cout << "Enter total cols for Array 2 " << endl;
```

```
cin >> colstwo;
```

```
int ** ptrone;
```

```
int ** ptrtwo;
```

```
Dma(ptrone,ptrtwo,rowsone,colsone,rowstwo,colstwo);
```

```
Adddata(ptrone,ptrtwo,rowsone,colsone,rowstwo,colstwo);
```

```
if(rowsone == rowstwo && colsone == colstwo &&  
rowstwo == colstwo)
```

```
{
```

```
Addition(ptrone,ptrtwo,rowsone,colsone,rowstwo,colstwo);
```

```
    Subtraction(ptrone, ptrtwo, rowsone, colsone, rowstwo,colstwo);
```

```
}
```

```
if(rowsone == colstwo)
```

```
{
```

```
Multiplication(ptrone,ptrtwo,rowsone,colsone,rowstwo,colstwo);
```

```
}
```

```
Freememory(ptrone,ptrtwo,rowsone,rowstwo);
```

```
return 0;
```

```
}
```



```
void Adddata(int ** ptrone, int ** ptrtwo, int
rowsone,int colsone,int rowstwo,int colstwo)
{

    for(int i =0; i < rowsone;i++)
    {
        for(int j =0; j < colsone; j++)
        {
            int num;
            cout << " Enter number for array one " << endl;
            cin >> num;
            ptrone[i][j] = num;

        }

    }

    for(int i =0; i < rowstwo;i++)
    {
        for(int j =0; j < colstwo; j++)
```

```

{
    int num;
    cout << " Enter number for array two " << endl;
    cin >> num;
    ptrtwo[i][j] = num;

}

}
}

void Dma(int **& ptrone, int **& ptrtwo, int rowsone,
int colsone,int rowstwo, int colstwo)

{
    ptrone = new int * [rowsone];
    for(int i =0; i < rowsone;i++)
    {
        ptrone[i] = new int[colsone];
    }
}

```

```
}
```

```
ptrtwo = new int * [rowstwo];
```

```
for(int u = 0; u < rowstwo;u++)
```

```
{
```

```
    ptrtwo[u] = new int[colstwo];
```

```
}
```

```
}
```

```
void Addition(int ** ptrone,int **ptrtwo, int rowsone,  
int colsone,int rowstwo, int colstwo)
```

```
{
```

```
    cout << "Addition answer is " << endl;
```

```
int Sum[rowsone][colsone];
```

```
for(int i =0; i < rowsone;i++)
```

```
{
```

```
    for(int j =0; j < colsone;j++)
```

```
{
```

```
    Sum[i][j] = ptrone[i][j] + ptrtwo[i][j];
```

```
    }  
}  
for(int i =0; i < rowsone;i++)  
{  
    for(int j =0; j < colsone;j++)  
    {  
        cout << " " << Sum[i][j];  
    }  
    cout << endl;  
}
```

```
}  
  
void Subtraction(int ** ptrone,int ** ptrtwo, int  
rowsone, int colsone,int rowstwo, int colstwo)  
{  
    cout << "Subtraction answer is " << endl;  
  
    int Subtract[rowsone][colsone];
```

```

for(int i =0; i < rowsone;i++)
{
    for(int j =0; j < colsone;j++)
    {
        Subtract[i][j] = ptrone[i][j] - ptrtwo[i][j];
    }
}

for(int i =0; i < rowsone;i++)
{
    for(int j =0; j < colsone;j++)
    {
        cout << " " << Subtract[i][j];
    }
    cout << endl;
}

}

void Multiplication(int ** ptrone,int ** ptrtwo, int
rowsone, int colsone,int rowstwo, int colstwo)

```

```
{  
    cout << " Multiplication is " << endl;  
    int Multiply[rowsone][colstwo];  
    for(int i =0; i < rowsone;i++)  
    {  
        for(int j =0; j < colstwo;j++)  
        {  
            Multiply[i][j] = 0;  
        }  
    }  
  
    for(int i =0; i < rowsone; i++)  
    {  
        for(int j = 0; j < colstwo; j++)  
        {  
            for(int k =0; k < colsone; k++)  
            {
```

```
        Multiply[i][j] = Multiply[i][j] + (ptrone[i][k] *  
ptrtwo[k][j]);
```

```
    }
```

```
    }
```

```
}
```

```
for(int i =0; i < rowsone;i++)
```

```
{
```

```
    for(int j =0; j < colstwo;j++)
```

```
    {
```

```
        cout << " " << Multiply[i][j];
```

```
    }
```

```
    cout << endl;
```

```
}
```

```
}
```

```
void Freememory(int ** ptrone, int ** ptrtwo,int  
rowsone, int rowstwo)  
{  
    for(int i =0; i < rowsone;i++)  
    {  
        delete ptrone[i];  
  
    }  
    delete [] ptrone;  
}
```

OUTPUT:


```
main.cpp
1 // 24K-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5 void Adddata(int ** ptrone, int ** ptrtwo, int rowsone, int colsone, int rowstwo, int colstwo
6 );
7 void Dma(int **& ptrone, int **& ptrtwo, int rowsone, int colsone, int rowstwo, int colstwo
8 );
9 void Addition(int ** ptrone, int ** ptrtwo, int rowsone, int colsone, int rowstwo, int
10 colstwo);
11 void Subtraction(int ** ptrone, int ** ptrtwo, int rowsone, int colsone, int rowstwo, int
12 colstwo);
13 void Multiplication(int ** ptrone, int ** ptrtwo, int rowsone, int colsone, int rowstwo, int
14 colstwo);
15 void Freememory(int ** ptrone, int ** ptrtwo, int rowsone, int rowstwo);
16
17 int main()
18 {
19     int rowsone;
20     cout << "Enter total rows for Array 1 " << endl;
21     cin >> rowsone;
22     int colsone;
23     cout << "Enter total cols for Array 1 " << endl;
24     cin >> colsone;
25     int rowstwo;
26     cout << "Enter total rows for Array 2 " << endl;
27     cin >> rowstwo;
28     int colstwo;
29     cout << "Enter total cols for Array 2 " << endl;
30     cin >> colstwo;
31
32     int ** ptrone;
33     int ** ptrtwo;
34     ptrone = new int*[rowsone];
35     ptrtwo = new int*[rowstwo];
36     for (int i = 0; i < rowsone; i++)
37     {
38         ptrone[i] = new int[colsone];
39     }
40     for (int i = 0; i < rowstwo; i++)
41     {
42         ptrtwo[i] = new int[colstwo];
43     }
44
45     Adddata(ptrone, ptrtwo, rowsone, colsone, rowstwo, colstwo);
46     Dma(ptrone, ptrtwo, rowsone, colsone, rowstwo, colstwo);
47     Addition(ptrone, ptrtwo, rowsone, colsone, rowstwo, colstwo);
48     Subtraction(ptrone, ptrtwo, rowsone, colsone, rowstwo, colstwo);
49     Multiplication(ptrone, ptrtwo, rowsone, colsone, rowstwo, colstwo);
50     Freememory(ptrone, ptrtwo, rowsone, rowstwo);
51 }
```

Output

```
Enter total rows for Array 1
2
Enter total cols for Array 1
2
Enter total rows for Array 2
2
Enter total cols for Array 2
2
Enter number for array one
1
Enter number for array one
2
Enter number for array one
3
Enter number for array one
4
Enter number for array two
5
Enter number for array two
6
Enter number for array two
7
Enter number for array two
8
Addition answer is
6 8
10 12
Subtraction answer is
-4 -4
-4 -4
Multiplication is
19 22
43 50
```

TASK#3).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
```

```
#include <iostream>
```

```
using namespace std;
```

```
struct Employee
```

```
{
```

string Employeeid;

string Name;

string Department;

double Salary;

};

**void Addinfo(Employee * employee, int
Totalemployees);**

**void Displayemployees(Employee * employee, int
Totalemployees);**

**void Searchemployee(Employee * employee, int
Totalemployees);**

**void Freememory(Employee * employee, int
Totalemployees);**

int main()

{

int Totalemployees;

```
cout << "Enter total employees " << endl;  
cin >> Totalemployees;
```

```
Employee * employee = new Employee  
[Totalemployees];  
Addinfo(employee, Totalemployees);  
Displayemployees(employee, Totalemployees);  
Searchemployee(employee, Totalemployees);  
Freememory(employee, Totalemployees);  
return 0;  
}  
  
void Addinfo(Employee * employee, int  
Totalemployees)  
{  
    for(int i =0; i < Totalemployees;i++)  
    {  
        string employeeid;  
        cout << " Enter employee id " << endl;  
        cin >> employeeid;
```

```
    string name;
    cout << " Enter employee name " << endl;
    cin >> name;
    string department;
    cout << " Enter employee department " << endl;
    cin >> department;
    double salary;
    cout << " Enter employee salary " << endl;
    cin >> salary;
    employee[i].Employeeid = employeeid;
    employee[i].Name = name;
    employee[i].Department = department;
    employee[i].Salary = salary;

}

}
```

```
void Displayemployees(Employee * employee, int
Totalemployees)
```

```

{
    for(int i =0; i < Totalemployees;i++)
    {
        cout << "Employeeid: " << employee[i].Employeeid
        << " " << "Employee name : " <<
        " " << employee[i].Name << " " << "Employee
        Department : " << employee[i].Department << " " <<
        "Employee salary : " << " " << employee[i].Salary <<
        endl;
    }
}

```

```

}

void Searchemployee(Employee * employee, int
Totalemployees)

```

```

{
    string EmployeeId;
    cout << " Enter employee id to search " << endl;
    cin >> EmployeeId;
    for(int i = 0; i < Totalemployees;i++)
    {

```

```

        if(employee[i].Employeeid == Employeeid)
        {
            cout << "Employee found " << endl;

            cout << "Employeeid: " <<
employee[i].Employeeid << " " << employee[i].Name <<
" " << employee[i].Department << " " <<
employee[i].Salary << endl;

        }
    }

}

void Freememory(Employee * employee, int
Totalemployees)
{
    delete [] employee;
}

```

OUTPUT:

```
main.cpp  Run  Output  Clear
1 // 24K-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5 struct Employee
6 {
7     string Employeeid;
8     string Name;
9     string Department;
10    double Salary;
11 };
12
13
14 void Addinfo(Employee * employee, int Totalemployees);
15 void Displayemployees(Employee * employee, int Totalemployees);
16 void Searchemployee(Employee * employee, int Totalemployees);
17 void Freememory(Employee * employee, int Totalemployees);
18
19 int main()
20 {
21
22     int Totalemployees;
23     cout << "Enter total employees " << endl;
24     cin >> Totalemployees;
25
26     Employee * employee = new Employee [Totalemployees];
27     Addinfo(employee, Totalemployees);
28     Displayemployees(employee, Totalemployees);
29     Searchemployee(employee, Totalemployees);
30     Freememory(employee, Totalemployees);
31     return 0;
32 }
33 void Addinfo(Employee * employee, int Totalemployees)
34 {
```

```
Enter total employees
2
Enter employee id
QWER8
Enter employee name
QASIM
Enter employee department
TECH
Enter employee salary
1290
Enter employee id
QWSD3
Enter employee name
ALI
Enter employee department
TECH
Enter employee salary
1290
Employeeid: QWER8 Employee name : QASIM Employee Department : TECH Employee salary : 1290
Employeeid: QWSD3 Employee name : ALI Employee Department : TECH Employee salary : 1290
Enter employee id to search
QWER8
Employee found
Employeeid: QWER8 QASIM TECH 1290

=== Code Execution Successful ===
```

TASK#4).

CODE:

// 24K-0559 BAZIL UDDIN KHAN

#include <iostream>

using namespace std;

struct Student

{

```
string Name;
int Rollnumber ;
int Marks[5];

};

void Addinfo(Student * student, int TotalStudents);
void Calculationanddisplay(Student * student, int
TTotalStudents,float * average );
void Freememory(Student * student, int
TotalStudents,float * average); // same

int main()
{

    int TotalStudents;
    cout << "Enter total students " << endl;
    cin >> TotalStudents;
    Student * student = new Student [TotalStudents];
```



```

float * average = new float [TotalStudents];
Addinfo(student,TotalStudents);
// DisplayResults(student,TotalStudents);
Calculationanddisplay(student, TotalStudents, average
);
//(student, TotalStudents,average);
Freememory(student, TotalStudents,average);
return 0;
}

void Addinfo(Student * student, int TotalStudents)
{
    for(int i =0; i < TotalStudents;i++)
    {
        string name;
        cout << " Enter employee name " << endl;
        cin >> name;
        int RollNumberR;
        cout << " Enter employee roll number : " << endl;
        cin >> RollNumberR;
    }
}

```

```
for(int i =0; i <5; i++)  
{  
    int Mark;  
    cout << " Enter Student Mark : " << endl;  
    cin >> Mark;  
    student[i].Marks[i] = Mark;  
}  
  
}  
}
```

```
void Calculationanddisplay(Student * student, int  
TotalStudents,float * average)
```

```
{
```

```
for(int i = 0; i < TotalStudents;i++)
```

```
{
```

```

int Totalmarks = 0;
for(int j =0; j <5; j++)
{
    Totalmarks = Totalmarks + student[i].Marks[i];

}
float Average = (Totalmarks/5);
average[i] = Average;
if(Average >= 90 )
{
    cout << "Your Grade is A+"<<" for student " <<
        " " << "Marks are : " <<average[i] << " " <<
i+1 << endl;
}
else if(Average >=80 && Average < 90)
{
    cout << "Your grade is A " << "for student " <<
" " << "Marks are : " << average[i] << " " << i+1 << endl;
}

```

```
else if(Average >= 70 && Average < 80)
{
    cout << "Your grade is B " <<"for student " <<"
    " << "Marks are " << average[i] <<
    " " << i+1 <<endl;
}
```

```
else if(Average >=60 && Average < 70)
{
    cout << "Your grade is C " << "for student " <<
    " " << "Marks are " << average[i] <<
    " " << i+1 << endl;
}
```

```
else if(Average >=50 && Average <60)
{
    cout << "Your grade is D " << "for student "
    << " " << " Marks are " << average[i] << " " << i+1 <<
    endl;
```

```

    }
    else
    {
        cout <<"Your grade is E " <<"for student " <<
" " << "Marks are " << average[i] << " " << i+1 << endl;
    }
}

```

```

}

void Freememory(Student * student, int
TotalStudents,float * average)
{
    delete [] student;
    delete [] average;
}

```

OUTPUT:

main.cpp	Run	Output
<pre>1 // 24K-0559 BAZIL UDDIN KHAN 2 #include <iostream> 3 using namespace std; 4 5 struct Student 6 { 7 string Name; 8 int Rollnumber ; 9 int Marks[5]; 10 }; 11 12 13 void Addinfo(Student * student, int TotalStudents); 14 void Calculationanddisplay(Student * student, int TTotalStudents,float * average); 15 void Freememory(Student * student, int TotalStudents,float * average); // same 16 17 int main() 18 { 19 20 int TotalStudents; 21 cout << "Enter total students " << endl; 22 cin >> TotalStudents; 23 Student * student = new Student [TotalStudents]; 24 float * average = new float [TotalStudents]; 25 Addinfo(student,TotalStudents); 26 // DisplayResults(student,TotalStudents); 27 Calculationanddisplay(student, TotalStudents, average); 28 //(student, TotalStudents,average); 29 Freememory(student, TotalStudents,average); 30 return 0; 31 } 32 void Addinfo(Student * student, int TotalStudents) 33 { 34 for(int i =0; i < TotalStudents;i++)</pre>		<pre>Enter total students 2 Enter employee name QASIM Enter employee roll number : 1234 Enter Student Mark : 10 Enter Student Mark : 45 Enter Student Mark : 60 Enter Student Mark : 30 Enter Student Mark : 79 Enter employee name ALI Enter employee roll number : 1905 Enter Student Mark : 18 Enter Student Mark : 40 Enter Student Mark : 39 Enter Student Mark : 90 Enter Student Mark : 55 Your grade is E for student Marks are 18 1 Your grade is E for student Marks are 40 2</pre>

TASK#5).

CODE:

// 24K-0559 BAZIL UDDIN KHAN

#include <iostream>

#include <string>

```
using namespace std;
```

```
void Dma(string *& ptr , string Stringone, string  
Stringtwo);
```

```
void Concatatestring(string * ptr, string Stringone,string  
Stringtwo);
```

```
void Comparestring(string * ptr,string Stringone, string  
Stringtwo);
```

```
void Lengthstring(string * ptr, string Stringone, string  
Stringtwo);
```

```
void Freememory(string * ptr);
```

```
int main()
```

```
{
```

```
    string Stringone;
```

```
    string Stringtwo;
```

```
    cout << "Enter string one " << endl;
```

```
    cin >> Stringone;
```

```
    cout << "Enter string two " << endl;
```

cin >> Stringtwo;

string * ptr;

Dma(ptr,Stringone,Stringtwo);

Concatatesting(ptr,Stringone, Stringtwo);

Comparestring(ptr, Stringone,Stringtwo);

Lengthstring(ptr,Stringone,Stringtwo);

Freememory(ptr);

return 0;

}

void Dma(string *& ptr , string Stringone, string Stringtwo)

{

ptr = new string[2];

ptr[0] = Stringone;

ptr[1] = Stringtwo;


```
}
```

```
void Concatatestring(string * ptr, string Stringone,string  
Stringtwo)
```

```
{
```

```
    cout << "The concatenated string is " <<  
(Stringone+Stringtwo) << endl;
```

```
}
```

```
void Comparestring(string * ptr,string Stringone, string  
Stringtwo)
```

```
{
```

```
    if(Stringone == Stringtwo)
```

```
    {
```

```
        cout << "Both string one : " << Stringone << " " << "  
and String two " << Stringtwo << " " << " is equal "  
<<endl;
```

```
    }
```

```
    else if(Stringone > Stringtwo)
```

```
    {
```

```
    cout << " string one : " << Stringone << " " << " is  
greater than String two " << Stringtwo <<endl;
```

```
}
```

```
else if(Stringone < Stringtwo)
```

```
{
```

```
    cout << " Stringtwo : " << Stringtwo << " " << " is  
greater than String one " << Stringone <<endl;
```

```
}
```

```
}
```

```
void Lengthstring(string * ptr, string Stringone, string  
Stringtwo)
```

```
{
```

```
    int Lengthone = 0;
```

```
    int Lengthtwo = 0;
```

```
    for(int i =0; Stringone[i] != '\0' ;i++)
```

```
{
```

```
    Lengthone++;  
}  
for(int i =0; Stringtwo[i] !='\0';i++)  
{  
    Lengthtwo++;  
}
```

```
    cout << "Length of string one is " << " " << Lengthone  
<< endl;
```

```
    cout << "Length of string two is " << " " << Lengthtwo  
<< endl;
```

```
}
```

```
void Freememory(string * ptr)
```

```
{
```

```
    delete [] ptr;
```

```
}
```

OUTPUT:

```
main.cpp  [Icons]  [Share]  [Run]  Output

1 // 24K-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 #include <string>
4 using namespace std;
5
6
7 void Dma(string *& ptr , string Stringone, string Stringtwo);
8 void Concatatestring(string * ptr, string Stringone,string Stringtwo);
9 void Comparestring(string * ptr,string Stringone, string Stringtwo);
10 void Lengthstring(string * ptr, string Stringone, string Stringtwo);
11 void Freememory(string * ptr);
12
13 int main()
14 {
15     string Stringone;
16     string Stringtwo;
17     cout << "Enter string one " << endl;
18     cin >> Stringone;
19     cout << "Enter string two " << endl;
20     cin >> Stringtwo;
21     string * ptr;
22     Dma(ptr,Stringone,Stringtwo);
23     Concatatestring(ptr,Stringone, Stringtwo);
24     Comparestring(ptr, Stringone,Stringtwo);
25     Lengthstring( ptr,Stringone,Stringtwo);
26     Freememory(ptr);
27
28
29
30
31     return 0;
32 }
33 void Dma(string *& ptr , string Stringone, string Stringtwo)
34 {
```

```
Enter string one
QASIM
Enter string two
ALI
The concatenated string is QASIMALI
string one : QASIM is greater than String two ALI
Length of string one is 5
Length of string two is 3

=== Code Execution Successful ===
```

TASK#6).

CODE:

// 24k-0559 BAZIL UDDIN KHAN

#include <iostream>

using namespace std;

struct Book

{

string Bookid;

```
string Title;
string Author;
string Isavailable;

};

void Addinfo(Book * book, int Totalbooks);
void Displaybooks(Book * book, int Totalbooks);
void Searchbooks(Book * book, int Totalbooks);
void Freememory(Book* book, int Totalbooks);
void Removebook(Book *& book, int & Totalbooks);

int main()
{
    cout << "Welcome to library Management system " <<
endl;

    int Totalbooks;
    cout << "Enter total Books " << endl;
```

```
cin >> Totalbooks;
```

```
Book * book = new Book [Totalbooks];
```

```
Addinfo(book,Totalbooks);
```

```
Displaybooks(book,Totalbooks);
```

```
Searchbooks(book, Totalbooks);
```

```
Removebook(book,Totalbooks);
```

```
Displaybooks(book,Totalbooks);
```

```
Freememory(book, Totalbooks);
```

```
return 0;
```

```
}
```

```
void Addinfo(Book * book, int Totalbooks)
```

```
{
```

```
for(int i =0; i < Totalbooks;i++)
```

```
{
```

```
    string bookid;
```

```
    cout << " Enter book id " << endl;
```

```
    cin >> bookid;
    string title;
    cout << " Enter book title " << endl;
    cin >> title;
    string author;
    cout << " Enter book author " << endl;
    cin >> author;
    string isAvailable;
    cout << " Enter in yes/no book status " << endl;
    cin >> isAvailable;
    book[i].Bookid = bookid;
    book[i].Title = title;
    book[i].Author = author;
    book[i].Isavailable = isAvailable;

}

}
```

```
void Displaybooks(Book * book, int Totalemployees)
```

```

{
    for(int i =0; i < Totalemployees;i++)
    {
        if(book[i].Isavailable == "yes")
        {
            cout << "Book id: " << book[i].Bookid << " " <<
"Book title : " << " " << book[i].Title << " " << "Book
Author is " << book[i].Author << " " << "book
availablity is : " << " " << book[i].Isavailable << endl;
        }
    }
}

```

```

}

void Searchbooks(Book * book, int Totalbooks)
{
    string bookId;
    cout << " Enter book id to search " << endl;
    cin >> bookId;
    for(int i = 0; i < Totalbooks;i++)

```



```

{
    if(book[i].Bookid == bookId)
    {
        cout << "Book found " << endl;

        cout << "Book id: " << book[i].Bookid << " " <<
book[i].Title << " " << book[i].Author << " " <<
book[i].Isavailable << endl;

    }
}

```

```

}

void Removebook(Book *& book, int & Totalbooks)
{
    int index = -1;
    string bookid;

    cout << "Enter book id whose informationistobe
removed " << endl;

    cin >> bookid;
}

```

```
for(int i =0; i < Totalbooks;i++)
{
    if(book[i].Bookid == bookid)
    {
        index = i;
    }
}
if(index == -1)
{
    cout << "Not found " << endl;

}
else
{
    Book * bok = new Book[Totalbooks - 1];
    int Index =0;
    for(int i = 0; i < Totalbooks;i++)
    {
        if(i != index)
```

```
{  
    bok[Index] = book[i];  
    Index++;  
}
```

```
}  
delete [] book;  
book = bok;
```

```
Totalbooks --;  
}
```

```
}  
void Freememory(Book * book, int Totalbooks)  
{  
    delete [] book;  
}
```

OUTPUT:

```
main.cpp  [ ] [ ] [ ] Share Run Output
1 // 24K-0559 BAZIL UDDIN KHAN
2
3 #include <iostream>
4 using namespace std;
5
6 struct Book
7 {
8     string Bookid;
9     string Title;
10    string Author;
11    string Isavailable;
12};
13
14
15 void Addinfo(Book * book, int Totalbooks);
16 void Displaybooks(Book * book, int Totalbooks);
17 void Searchbooks(Book * book, int Totalbooks);
18 void Freememory(Book* book, int Totalbooks);
19 void Removebook(Book *& book, int & Totalbooks);
20
21 int main()
22 {
23     cout << "Welcome to library Management system " << endl;
24
25     int Totalbooks;
26     cout << "Enter total Books " << endl;
27     cin >> Totalbooks;
28
29     Book * book = new Book [Totalbooks];
30     Addinfo(book, Totalbooks);
31     Displaybooks(book, Totalbooks);
32     Searchbooks(book, Totalbooks);
33
34     Removebook(book, Totalbooks);
35 }
```

```

Welcome to library Management system
Enter total Books
3
Enter book id
age
Enter book title
fguo
Enter book author
fgu
Enter in yes/no book status
yes
Enter book id
vgju
Enter book title
6rfr
Enter book author
kcf
Enter in yes/no book status
no
Enter book id
yfhc
Enter book title
fuuf
Enter book author
vk
Enter in yes/no book status
yes
Book id: age Book title : fguo Book Author is fgu book availability is : yes
Book id: yfhc Book title : fuuf Book Author is vk book availability is : yes
Enter book id to search
yfhc
Book found
Book id: yfhc fuuf vk yes
Enter book id whose information is to be removed
```

```
main.cpp  [ ] [ ] [ ] Share Run Output
1 // 24K-0559 BAZIL UDDIN KHAN
2
3 #include <iostream>
4 using namespace std;
5
6 struct Book
7 {
8     string Bookid;
9     string Title;
10    string Author;
11    string Isavailable;
12};
13
14
15 void Addinfo(Book * book, int Totalbooks);
16 void Displaybooks(Book * book, int Totalbooks);
17 void Searchbooks(Book * book, int Totalbooks);
18 void Freememory(Book* book, int Totalbooks);
19 void Removebook(Book *& book, int & Totalbooks);
20
21 int main()
22 {
23     cout << "Welcome to library Management system " << endl;
24
25     int Totalbooks;
26     cout << "Enter total Books " << endl;
27     cin >> Totalbooks;
28
29     Book * book = new Book [Totalbooks];
30     Addinfo(book, Totalbooks);
31     Displaybooks(book, Totalbooks);
32     Searchbooks(book, Totalbooks);
33
34     Removebook(book, Totalbooks);
35 }
```

```

age
Enter book title
fguo
Enter book author
fgu
Enter in yes/no book status
yes
Enter book id
vgju
Enter book title
6rfr
Enter book author
kcf
Enter in yes/no book status
no
Enter book id
yfhc
Enter book title
fuuf
Enter book author
vk
Enter in yes/no book status
yes
Book id: age Book title : fguo Book Author is fgu book availability is : yes
Book id: yfhc Book title : fuuf Book Author is vk book availability is : yes
Enter book id to search
yfhc
Book found
Book id: yfhc fuuf vk yes
Enter book id whose information is to be removed
vgju
Book id: age Book title : fguo Book Author is fgu book availability is : yes
Book id: yfhc Book title : fuuf Book Author is vk book availability is : yes
```

TASK#7).

CODE:

// 24k-0559 BAZIL UDDIN KHAN

#include <iostream>

using namespace std;

void Inputvalues(int * ptr, int Totalelements);

void Swapnum(int *x, int * y);

void Freememory(int *ptr);

void Reversearray(int *& ptr, int Totalelements);

void Display(int * ptr, int Totalelements);

int main()

{

int x,y;

cout << "Enter x " << endl;

cin >>(x);

cout << "Enter y " << endl;

cin >> (y);

```
int Totalelements;
cout << "Enter total elements " << endl;
cin >> Totalelements;

cout << "x value before sawp is " << x << " and y value
before sawp " << y << endl;

Swapnum(&x,&y);

int * ptr = new int[Totalelements];
Inputvalues(ptr,Totalelements);
//Swapnum(&x, & y);
Reversearray( ptr, Totalelements);
Display( ptr, Totalelements);
Freememory(ptr);

cout << endl;

cout <<"xis now : " << (x)<< " and y is now " << (y) <<
endl;

return 0;
}
```

```
void Inputvalues(int * ptr, int Totalelements)
{
    for(int i =0; i < Totalelements;i++)
    {
        int num;
        cout << "Enter num "<<endl;
        cin >> num;
        ptr[i] = num;
    }
}
```

```
void Swapnum(int* x, int* y)
{
    int Temp = *x;
    *x = *y;
    *y = Temp;
```

```
}
```

```
void Freememory(int *ptr)
```

```
{
```

```
    delete [] ptr;
```

```
}
```

```
void Reversearray(int *& ptr, int Totalelements)
```

```
{
```

```
    int * Array= new int [Totalelements];
```

```
    // for(int j =0; j < Totalelements;j++)
```

```
    // {
```

```
        // Array[j] = ptr[j];
```

```
    // }
```

```
    int Index = Totalelements-1;
```

```
    for(int i =0;i < Totalelements; i++)
```

```
    {
```

```
        Array[Index] = ptr[i];
```

```
        Index--;
```



```
}
```

```
for(int i = 0;i < Totalelements;i++)
```

```
{
```

```
    (ptr[i]) = (Array[i]);
```

```
}
```

```
delete [] Array;
```

```
}
```

```
void Display(int * ptr, int Totalelements)
```

```
{
```

```
    for(int i = 0; i < Totalelements;i++)
```

```
    {
```

```
        // cout << "Reverse order num is " ;
```

```
        cout <<" " <<ptr[i];
```

```
    }
```

```
}
```

OUTPUT:

```
main.cpp  Run  Share  Output
1 // 24k-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5 void Inputvalues(int * ptr, int Totalelements);
6 void Swapnum(int *x, int * y);
7 void Freememory(int *ptr);
8 void Reversearray(int *& ptr, int Totalelements);
9 void Display(int * ptr, int Totalelements);
10
11 int main()
12 {
13     int x,y;
14     cout << "Enter x " << endl;
15     cin >> ( x );
16     cout << "Enter y " << endl;
17     cin >> (y);
18
19     int Totalelements;
20     cout << "Enter total elements " << endl;
21     cin >> Totalelements;
22     cout << "x value before sawp is " << x << " and y value before sawp " << y << endl;
23     Swapnum(&x,&y);
24     int * ptr = new int[Totalelements];
25     Inputvalues(ptr,Totalelements);
26     //Swapnum(&x, &y);
27     Reversearray( ptr, Totalelements);
28     Display( ptr, Totalelements);
29     Freememory(ptr);
30     cout << endl;
31     cout << "xis now : " << (x)<< " and y is now " << (y) << endl;
32
33     return 0;
34 }
```

```
Output
Enter x
12
Enter y
78
Enter total elements
3
x value before sawp is 12 and y value before sawp 78
Enter num
12
Enter num
13
Enter num
14
14 13 12
xis now : 78 and y is now 12

=== Code Execution Successful ===
```

TASK#8).

CODE: // 24k-0559 BAZIL UDDIN KHAN

```
#include <iostream>
```

```
using namespace std;
```

```
struct Product
```

```
{
```

```
    string Productid;
```

```
    string Name;
    int Quantity;
    double Price;

};

void Addinfo(Product * product, int Totalproducts);
void DisplayTotal(Product * product, int Totalproducts);
void Removeproduct(Product*& product, int
&Totalproducts);
void Updateproducts(Product *& product,int
Totalproducts);
void Freememory(Product * product, int Totalproducts);
void Display(Product * product, int Totalproducts);

int main()
{

    int Totalproducts;
```

```
cout << "Enter total products " << endl;
```

```
cin >> Totalproducts;
```

```
Product * product = new Product [Totalproducts];
```

```
Addinfo(product,Totalproducts);
```

```
DisplayTotal(product,Totalproducts);
```

```
Removeproduct(product,Totalproducts);
```

```
Updateproducts(product,Totalproducts);
```

```
Display( product, Totalproducts);
```

```
Freememory(product,Totalproducts);
```

```
return 0;
```

```
}
```

```
void Addinfo(Product * product, int Totalproducts)
```

```
{
```

```
for(int i =0; i < Totalproducts;i++)
```

```
{
```

```
    string productid;
```

```
    cout << " Enter product id " << endl;
```

```
    cin >> productid;
```

```
    string name;
    cout << " Enter product name " << endl;
    cin >> name;
    int quantity;
    cout << " Enter priduct total quantity  " << endl;
    cin >> quantity;
    double price;
    cout << " Enter total products  of product " <<
endl;
    cin >> price;
    product[i].Productid = productid;
    product[i].Name = name;
    product[i].Quantity = quantity;
    product[i].Price = price;

}
}
void DisplayTotal(Product * product, int Totalproducts)
{
```

```

double Total =0;
for(int i =0; i < Totalproducts;i++)
{
    Total = Total + (product[i].Quantity *
product[i].Price);
}

cout << " The total price of the prsent " << Total <<
endl;

}

```

```

void Removeproduct(Product*& product, int &
Totalproducts)
{
    int index = -1;
    string productid;
    cout <<" Enter product id whose informationistobe
removed " << endl;
    cin >> productid;
    for(int i =0; i < Totalproducts;i++)

```

```
{  
    if(product[i].Productid == productid)  
    {  
        index = i;  
    }  
}  
if(index == -1)  
{  
    cout << "Not found " << endl;  
  
}  
else  
{  
    Product * pro = new Product[Totalproducts - 1];  
    int Index =0;  
    for(int i = 0; i < Totalproducts;i++)  
    {  
        if(i != index)  
        {
```

```
        pro[Index] = product[i];  
        Index++;  
    }
```

```
    }  
    delete [] product;  
    product = pro;  
    Totalproducts --;
```

```
}
```

```
}
```

```
void Updateproducts(Product *& product,int  
Totalproducts)
```

```
{
```

```
    string productid;
```



```
cout << "Enter product id whose information to be updated " << endl;
```

```
cin >> productid;
```

```
for(int i =0; i < Totalproducts;i++)
```

```
{
```

```
    if(product[i].Productid == productid)
```

```
    {
```

```
        string Newproductid;
```

```
        cout << "Enter updated product id " << endl;
```

```
        cin >> Newproductid;
```

```
        string Newname;
```

```
        cout << "Enter updated name " << endl;
```

```
        cin >> Newname;
```

```
        int quantity;
```

```
        cout << "Enter updated Quantity " << endl;
```

```
        cin >> quantity;
```

```
        double price;
```

```
        cout << "Enter updated price " << endl;
```

```
        cin >> price;
```

```
product[i].Productid = Newproductid;
```

```
product[i].Name = Newname;
```

```
product[i].Quantity = quantity;
```

```
product[i].Price = price;
```

```
cout << endl;
```

```
cout << "Successfully updated " << endl;
```

```
}
```

```
}
```

```
}
```

```
void Display(Product * product, int Totalproducts)
```

```
{
```

```
    for(int i =0; i < Totalproducts;i++)
```

```
    {
```

```
        cout << "Product id is :" << product[i].Productid <<  
endl;
```

```
        cout << "Product name is :" << product[i].Name << endl;
```

```
        cout << "Product quantity is :" << product[i].Quantity << endl;
```

```
        cout << "Product price is :" << product[i].Price << endl;
```

```
    }
```

```
}
```

```
void Freememory(Product * product, int Totalproducts)
```

```
{
```

```
    delete [] product;
```

```
}
```

OUTPUT:

main.cpp	Output
<pre>1 // 24k-0559 BAZIL UDDIN KHAN 2 3 #include <iostream> 4 using namespace std; 5 6 struct Product 7 { 8 string Productid; 9 string Name; 10 int Quantity; 11 double Price; 12 }; 13 14 void Addinfo(Product * product, int Totalproducts); 15 void DisplayTotal(Product * product, int Totalproducts); 16 void Removeproduct(Product*& product, int &Totalproducts); 17 void Updateproducts(Product *& product,int Totalproducts); 18 void Freememory(Product * product, int Totalproducts); 19 void Display(Product * product, int Totalproducts); 20 21 int main() 22 { 23 int Totalproducts; 24 cout << "Enter total products " << endl; 25 cin >> Totalproducts; 26 27 Product * product = new Product [Totalproducts]; 28 Addinfo(product,Totalproducts); 29 DisplayTotal(product,Totalproducts); 30 Removeproduct(product,Totalproducts); 31 Updateproducts(product,Totalproducts); 32 Freememory(product,Totalproducts); 33 Display(product,Totalproducts); 34 }</pre>	<pre>Enter total products 3 Enter product id eseti7 Enter product name yfyfy Enter prduct total quantity 5 Enter total products of product 100 Enter product id ert0 Enter product name tyuio Enter prduct total quantity 6 Enter total products of product 300 Enter product id qwe Enter product name qrty Enter prduct total quantity 10 Enter total products of product 100 The total price of the present 3300 Enter product id whose informationistobe removed qwe Enter product id whose information to be updated ert0 Enter updated product id ert1 Enter undated name</pre>

main.cpp	Output
<pre>1 // 24k-0559 BAZIL UDDIN KHAN 2 3 #include <iostream> 4 using namespace std; 5 6 struct Product 7 { 8 string Productid; 9 string Name; 10 int Quantity; 11 double Price; 12 }; 13 14 void Addinfo(Product * product, int Totalproducts); 15 void DisplayTotal(Product * product, int Totalproducts); 16 void Removeproduct(Product*& product, int &Totalproducts); 17 void Updateproducts(Product *& product,int Totalproducts); 18 void Freememory(Product * product, int Totalproducts); 19 void Display(Product * product, int Totalproducts); 20 21 int main() 22 { 23 int Totalproducts; 24 cout << "Enter total products " << endl; 25 cin >> Totalproducts; 26 27 Product * product = new Product [Totalproducts]; 28 Addinfo(product,Totalproducts); 29 DisplayTotal(product,Totalproducts); 30 Removeproduct(product,Totalproducts); 31 Updateproducts(product,Totalproducts); 32 Freememory(product,Totalproducts); 33 Display(product,Totalproducts); 34 }</pre>	<pre>Enter total products of product 300 Enter product id qwe Enter product name qrty Enter prduct total quantity 10 Enter total products of product 100 The total price of the present 3300 Enter product id whose informationistobe removed qwe Enter product id whose information to be updated ert0 Enter updated product id ert1 Enter updated name ft Enter updated Quantity 5 Enter updated price 109 Successfully updated Product id is :eseti7 Product name is :yfyfy Product quantity is :5 Product price is :100 Product id is :ert1 Product name is :ft Product quantity is :5 Product price is :109</pre>

TASK#9)

CODE:

```
// 24k-0559 BAZIL UDDIN KHAN
```

```
#include <iostream>
```

```
using namespace std;
```

```
struct Student
```

```
{
```

```
    string Name;
```

```
    int Rollnumber ;
```

```
    int * Marks;
```

```
};
```

```
void Addinfo(Student * student, int TotalStudents,int  
Totalsubjarr[]);
```

```
void Displayhighest(Student * student, int  
TotalStudents,int * Total, int Totalsubjarr[]);
```

```
void Freememory(Student * student, int  
TotalStudents,int * Total);
```

```
int main()
{

    int TotalStudents;
    cout << "Enter total students " << endl;
    cin >> TotalStudents;
    Student * student = new Student [TotalStudents];
    int * Total = new int[TotalStudents];
    int totalsubjarr[50] ;

    Addinfo(student,TotalStudents,totalsubjarr);

    Displayhighest(student, TotalStudents,Total
,totalsubjarr);

    Freememory(student, TotalStudents,Total);
    return 0;
}

void Addinfo(Student * student, int TotalStudents,int
Totalsubjarr[])
```

```
{  
    int Totalsubj = 0;  
    for(int i =0; i < TotalStudents;i++)  
    {  
  
        string name;  
        cout << " Enter Student name " << endl;  
        cin >> name;  
        student[i].Name = name;  
        int RollNumberR;  
        cout << " Student roll number : " << endl;  
        cin >> RollNumberR;  
        student[i].Rollnumber = RollNumberR;  
  
        int Entertotalmarks;  
        cout << " enter total subjectS marks " << " for  
student " << i+1 << endl;  
        cin >> Entertotalmarks;
```

```
    student[i].Marks= new int [Entertotalmarks];  
    for(int j =0; j <Entertotalmarks; j++)  
    {  
        int Mark;  
        cout << " Enter Student Mark : " << endl;  
        cin >> Mark;  
        student[i].Marks[j] = Mark;  
  
    }  
    Totalsubjarr[Totalsubj] = Entertotalmarks;  
    Totalsubj++;  
  
}  
}
```

```
void Displayhighest(Student * student, int  
TotalStudents,int * Total,int Totalsubjarr[])  
{
```



```
for(int i = 0; i < TotalStudents;i++)  
{  
    int Totalmarks = 0;  
    for(int j =0; j <Totalsubjarr[i]; j++)  
    {  
        Totalmarks = Totalmarks + student[i].Marks[j];  
    }  
}
```

```
Total[i] = Totalmarks;
```

```
}
```

```
int Highestmarks = Total[0];  
int index = 0;  
for(int i = 0; i < TotalStudents;i++)  
{  
    if(Highestmarks < Total[i])
```

```
{  
    Highestmarks = Total[i];  
    index = i;  
}  
}
```

```
cout << "The Student Name is " <<  
student[index].Name << " " << " and the roll number is "  
<< student[index].Rollnumber << " " << " and total  
marks are " << Highestmarks << endl;
```

```
}  
  
void Freememory(Student * student, int  
TotalStudents,int * Total)  
{  
    delete [] student;  
    delete [] Total;  
}
```

OUTPUT:

main.cpp	Output
<pre>1 // 24k-0559 BAZIL UDDIN KHAN 2 #include <iostream> 3 using namespace std; 4 5 struct Student 6 { 7 string Name; 8 int Rollnumber ; 9 int * Marks; 10 } 11 }; 12 13 void Addinfo(Student * student, int TotalStudents,int Totalsubjarr[]); 14 void Displayhighest(Student * student, int TotalStudents,int * Total, int Totalsubjarr[]) 15); 16 void Freememory(Student * student, int TotalStudents,int * Total); 17 18 int main() 19 { 20 int TotalStudents; 21 cout << "Enter total students " << endl; 22 cin >> TotalStudents; 23 Student * student = new Student [TotalStudents]; 24 int * Total = new int[TotalStudents]; 25 int totalsubjarr[50] ; 26 27 Addinfo(student,TotalStudents,totalsubjarr); 28 Displayhighest(student, TotalStudents,Total ,totalsubjarr); 29 30 Freememory(student, TotalStudents,Total); 31 return 0; 32 } 33 void Addinfo(Student * student, int TotalStudents,int Totalsubjarr[])</pre>	<pre>Enter total students 3 Enter Student name QASIM Student roll number : 123 enter total subjectS marks for student 1 3 Enter Student Mark : 12 Enter Student Mark : 10 Enter Student Mark : 8 Enter Student name ALI Student roll number : 345 enter total subjectS marks for student 2 3 Enter Student Mark : 12 Enter Student Mark : 56 Enter Student Mark : 67 Enter Student name qas Student roll number : 123 enter total subjectS marks for student 3 2 Enter Student Mark : 12</pre>

```
main.cpp 1 // 24k-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5 struct Student
6 {
7     string Name;
8     int Rollnumber ;
9     int * Marks;
10 }
11
12 void Addinfo(Student * student, int TotalStudents,int Totalsubjarr[]);
13 void Displayhighest(Student * student, int TotalStudents,int * Total, int Totalsubjarr[]);
14 void Freememory(Student * student, int TotalStudents,int * Total);
15
16 int main()
17 {
18     int TotalStudents;
19     cout << "Enter total students " << endl;
20     cin >> TotalStudents;
21     Student * student = new Student [TotalStudents];
22     int * Total = new int[TotalStudents];
23     int totalsubjarr[50] ;
24
25     Addinfo(student,TotalStudents,totalsubjarr);
26     Displayhighest(student, TotalStudents,Total ,totalsubjarr);
27     Freememory(student, TotalStudents,Total);
28     return 0;
29 }
```

Output

```
Student roll number :
123
enter total subjectS marks for student 1
3
Enter Student Mark :
12
Enter Student Mark :
10
Enter Student Mark :
8
Enter Student name
ALI
Student roll number :
345
enter total subjectS marks for student 2
3
Enter Student Mark :
12
Enter Student Mark :
56
Enter Student Mark :
67
Enter Student name
qas
Student roll number :
123
enter total subjectS marks for student 3
2
Enter Student Mark :
12
Enter Student Mark :
8
The Student Name is ALI and the roll number is 345 and total marks are 135
```

TASK#10).

CODE:

```
// 24K-0559 BAZIL UDDIN KHAN
```

```
#include <iostream>
```

```
using namespace std;
```

```
void Adddata(int ** ptr, int rows,int cols);
```

```
void Dma(int **& ptr, int rows, int cols);
```

```
void Displayinfo(int ** ptr, int rows, int cols);
```

```
void Freememory(int ** ptr, int rows);
```

```
int main()
```

```
{
```

```
    int rows;
```

```
    cout << "Enter total rows " << endl;
```

```
    cin >> rows;
```

```
    int cols;
```

```
    cout << "Enter total cols " << endl;
```

```
    cin >> cols;
```

```
    int ** ptr;
```

```
    Dma(ptr, rows,cols);
```

```
    Adddata(ptr,rows,cols);
```

```
    Displayinfo(ptr,rows,cols);
```

```
Freememory(ptr,rows);
```

```
return 0;
```

```
}
```

```
void Adddata(int ** ptr, int rows, int cols)
```

```
{
```

```
for(int i =0; i < rows;i++)
```

```
{
```

```
for(int j =0; j < cols; j++)
```

```
{
```

```
int num;
```

```
cout << " Enter number " << endl;
```

```
cin >> num;
```

```
ptr[i][j] = num;
```

```
}
```

```
}  
}
```

```
void Dma(int **& ptr, int rows, int cols)
```

```
{  
    ptr = new int * [rows];  
    for(int i =0; i < rows;i++)  
    {  
        ptr[i] = new int[cols];  
    }  
}
```

```
void Displayinfo(int ** ptr, int rows, int cols)
```

```
{  
    cout << "Original Array is " << endl;  
    for(int i = 0; i < rows;i++)  
    {  
        for(int j =0; j < cols;j++)  
        {
```

```
        cout << " " << ptr[i][j];  
    }  
    cout << endl;  
}
```

```
cout << "Transpose array is " << endl;  
int Newarray[rows][cols];  
int index =0;  
for(int i =0; i < rows;i++)  
{
```

```
    for(int j = 0; j < cols;j++)  
    {  
        Newarray[j][i] = ptr[i][j];  
    }  
}
```

```
for(int i = 0; i < rows;i++)  
{
```



```
        for(int j =0; j < cols;j++)
        {
            cout << " " << Newarray[i][j];

        }
        cout << endl;
    }

}
```

```
void Freememory(int ** ptr, int rows)
{
    for(int i =0; i < rows;i++)
    {
        delete ptr[i];
    }
    delete [] ptr;
}
```

OUTPUT:

main.cpp



Share

Run

Output

```
1 // 24K-0559 BAZIL UDDIN KHAN
2 #include <iostream>
3 using namespace std;
4
5 void Adddata(int ** ptr, int rows, int cols);
6
7 void Dma(int **& ptr, int rows, int cols);
8 void Displayinfo(int ** ptr, int rows, int cols);
9
10 void Freememory(int ** ptr, int rows);
11
12
13 int main()
14 {
15     int rows;
16     cout << "Enter total rows " << endl;
17     cin >> rows;
18     int cols;
19     cout << "Enter total cols " << endl;
20     cin >> cols;
21
22     int ** ptr;
23
24     Dma(ptr, rows, cols);
25     Adddata(ptr, rows, cols);
26     Displayinfo(ptr, rows, cols);
27
28     Freememory(ptr, rows);
29
30     return 0;
31 }
32 void Adddata(int ** ptr, int rows, int cols)
33 {
34
```

```
Enter total rows
2
Enter total cols
2
Enter number
1
Enter number
2
Enter number
3
Enter number
4
Original Array is
1 2
3 4
Transpose array is
1 3
2 4

=== Code Execution Successful ===
```